TESTING AND INVESTIGATIONS OF REACTION WHEELS

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Abstract. The first part of the paper presents the results of measurements on torque and force irregularities of running reaction wheels. Six different, current technology European wheels, both with ball- and magnet bearings, and with d.c. and a.c. drive motors are examined. Time-domain behaviour around zero wheel speed as well as noise at certain constant wheel speeds is investigated.
The second part of the paper discusses the performance aspects of the operation of reaction wheel systems having their own internal feedback control system. The reduction in wheel induced spacecraft attitude noise by employing wheel velocity or wheel attitude feedback is analysed in the continuous time domain.

Keywords. Attitude control;actuators; reaction wheels; model following control; velocity control; controllers; ball bearings, magnetic bearings.

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INTRODUCTION

High performance requirements in attitude pointing for current and future satellite missions have reached the point where torque- and force irregularities of a random or deterministic nature in reaction wheels become noticeable in pointing error budgets.
The increasing power of on-board processors allows more freedom in the design and operation of wheel control systems, and makes a review of earlier trade-offs desirable.
These two considerations have led to the present investigation, whose goal is to enable attitude control designers:

- to make rational predictions on pointing performance limitations imposed by reaction wheel imperfections. This is usually needed in feasibility studies or in early mission explorations.
- to obtain guidelines on how to configure reaction wheels and on how to operate them best from a systems viewpoint. This means considering appropriate modes that can be expected to occur in a single satellite mission.

This paper reports on an experimental and theoretical investigation of these aspects. The first part of the paper contains a discussion of the test method and the results of tests, executed on three different ball-bearing wheels and on three different wheels with magnetic bearings. All wheels are of European design. In the second part wheel control methods to minimize the effect of torque irregularities on spacecraft attitude motions are analysed.

MEASUREMENTS AND STATISTICAL ANALYSIS OF FORCE AND TORQUE IRREGULARITIES

Reaction Wheels Tested

Two classes of reaction wheels with respect to bearing type were tested, viz.: wheels with ball bearings and wheels with magnetic bearings. A summary of the different wheels is shown in Table 1.

Ball-bearing wheels. The ball-bearing wheels all contain the same type of bearing unit of special design developed by Teldix.
This ball-bearing system consists of a pair of preloaded angular contact ball bearings of relatively large dimensions. The bearings for the different type of wheels are exactly the same, only the distance between the bearings depends on the size of the wheel.
The drive motor for the d.c. wheels (TD and TN) is an ironless and brushless motor capable of speed reversals. Reaction wheel TA is driven by an asynchronous two phase squirrel-cage a.c. induction motor. The windings are mounted on a laminated iron core forming an 8 pole-pair stator. The power supply to this motor is a two-phase 400 Hz square wave with

90° mutual phase shift, resulting in a synchronous speed of 3000 rpm.
Figure 1 gives an impression of the construction of wheel TN.

Magnetic-bearing wheels. In a magnetic-bearing reaction wheel the rotor is positioned with respect to the stator by magnetic forces in an entirely contactless manner. Complete positional and orientational control of the rotor requires in principle the control of six degrees of freedom (DOF): three translations and three rotations. One degree of rotational motion is controlled by the drive motor and is not part of the controlling task of the magnetic bearing. Therefore the magnetic bearing must control five degrees of freedom.
This can be performed by active elements (electromagnets varying the excitation as a function of rotor position and velocity) or by passive elements (permanent magnets or electromagnets with fixed excitation). It is physically impossible to constrain all five degrees of freedom by passive magnetic elements; active control of at least one degree of freedom is necessary.

Wheel SM applies a 1-DOF active magnetic bearing, in which the translational position of the rotor along its axis of rotation is the only actively controlled DOF.
Wheels EM and SM employ a 2-DOF active magnetic bearing, in which the two translational degrees of freedom of the rotor orthogonal to the rotation axis are actively controlled.
The drive motors of all magnetic bearing wheels are ironless and brushless d.c. motors. Commutation is accomplished electronically.

Measurement Set-Up

The wheel under test was mounted on a dynamometer placed on a stable test table under a vacuum jar with its axis of rotation in vertical direction. See Fig. 2. The vacuum jar was used because the tested wheels normally operate in vacuum. Besides that the vacuum suppressed the transfer of acoustic noise from the testroom to the dynamometer. This device is a piezo electric dynamometer (Kistler type 9273) which delivered axial torque and force (M_z and F_z respectively) and two radial force components (F_x and F_y).

The charge outputs of the sensor were amplified (integrated) in charge amplifiers to force and torque components, low-pass antialiasing filtered and fed to the analyser which computed through a Fast-Fourier transform the power spectrum. The results were presented on the analyser oscilloscope screen: they also could be plotted on a digital plotter and stored on cassette tape in the desk-top computer. This enabled at any time afterwards the exact reproduction of each plot; besides that also the execution of interim computations on the spectra were possible.

For some special runs, i.e. wheel drive motor power-off runs, a direct coupling of the analysers time base (and thus the sampling frequency) to the wheel speed was necessary. This was performed through the spectrum tracking adapter, using the wheel speed (tachometer) signal. This mode of operation is called "order" mode. When in this situation the wheel speed varied, the analysers sampling frequency and maximum plot frequency varied proportionally. In such a situation it was also necessary to change in a similar way the crossoverfrequency of the antialiasing low-pass filter. This was performed by the tracking filter.

Test Programme

Resonance search run. The combination of the dynamometer and its load mass constituted a spring-mass system having a fundamental resonance. In order to avoid wheel speeds to close to this resonance each test series was started with a resonances search run. This search run was executed by a spin-up from zero to maximum wheel speed at constant drive motor current value. During spin-up the charge amplifier output amplitude and the wheel speed value were plotted on an analog x-y recorder. This plot shows that the output amplitude reaches larger values at certain wheel speeds. This is caused by the fact that at these wheel speeds certain peaks in the torque and force power spectrum of the wheel (which are all harmonics of the wheel speed)coincide with the fundamental resonance of the spring-mass system. An example is shown in Fig. 3. By comparison of the frequencies of the peaks it is possible to determine which harmonic excites the fundamental resonance and the latter's value (432.5 Hz in Fig. 3).

In the measured spectra discussed below a linear distorsion may occur due to the transfer function associated with the spring-mass system mentioned above. Indeed in some cases, due to large values of wheel mass or wheel housing inertia, the lowest resonance frequency was within the frequency range of interest, resulting in too high values of the measured wheel noise spectra at frequencies in the neighbourhood of such a resonance frequency.

By the impulsive input method the sensor-wheel general transfer function shape was measured. As this method only is qualitative the resulting values were not used to correct the measurements, but they may be used in the study of the spectra.
This method consists of excitation of the wheel-dynamometer system manually by a series of small impulses in the proper component direction (M_z, F_z or F_x).

Start-up run. A number of start-up runs were made on the M_z channel, for different constant spin motor current values. The output plot of such a run represents the wheel acceleration torque step response, on which torque irregularities are superimposed which in case of a d.c. drive motor are mainly commutation peaks.
From the resulting plots one can get an impression of the irregularities caused by the

drive motor. Examples are shown in the Figs. 4 and 5.

Speed reversals and run-downs. These measurements are also executed on the M_z channel only. They both give an impression of the Coulomb friction torque. A speed-reversal starts from a running-wheel situation in which a constant motor-current is switched to the opposite direction. This causes the wheel to decelerate to zero wheel speed after which it accelerates in the opposite direction. The plot in question shows the Coulomb friction to change over twice its value at the zero wheel speed moment. See Fig. 6.

In a run-down test with motor power off, the torque during the last few seconds before wheel standstill is measured. The resulting plots also give an indication of the wheel Coulomb friction which falls back to zero at the wheel stop moment. An example for wheel TN is shown in Fig. 7. The measured Coulomb friction agrees rather well with the value of Fig. 6.

Spectral runs. The frequency range of interest was from 0.02 Hz to 500 Hz. As relatively sharp spectral lines constitute the main contribution to the spectra, the determination of their location and magnitude was a prime concern. For this reason most spectral measurements were executed at constant values of the wheel speed. The analyser's maximum range was selected to be 5 Hz and 500 Hz. As the spectral computations resulted in 400 frequency elements, the frequency resolutions are 0.0125 Hz and 1.25 Hz respectively. The goal for the statistical accuracy of the measurements was 25%. The standard error for spectral density estimates follows from

$$\varepsilon^2 = 1/\Delta f T_r = f_s/\Delta f N$$

with Δf the frequency resolution, f_s the sampling frequency, T_r the record duration, N the total sample size.
For a standard error ε = 25% the sample size should be: N = 16384. In the Nicolet spectrum averaging SUM mode at 32 integration periods a sample size of N = 16896 data points was used to compute an averaged spectrum.

As in ball bearings the friction values have a considerable temperature dependence, constant wheel speeds for the current controlled wheels could only be obtained after a long time of stabilization. Different values of constant wheel speed avoiding resonance were selected ranging from low to high values within the possible range; one of the values was selected in negative direction of rotation. These wheel speeds are indicated in Table 2.

Besides the spectral runs as mentioned above, a number of special runs were executed.

The spectral measurements on the wheels running at varying speed with the drive motor switched-off provide a means to distinguish some types of wheel speed dependent noise such as commutation noise from wheel speed independent noise sources such as for instance mains noise. These runs were executed in the analyser's "order" mode.

In order to determine the threshold of the measurement system eigennoise measurements were executed in the nominal set-up with wheels at standstill. For the magnetic bearing wheels these measurements were executed with wheel suspension on.

Some Results

As an illustration this chapter contains a selection of some of the more important results. A more comprehensive treatment is given by Bosgra and Smilde (1982).

Stiction and friction determination. Wheel behaviour in the near zero speed region also is of importance when speed reversals pass through the stiction and friction irregularities. A simplified friction model is shown in Fig. 8. The stiction values of the wheels were computed from the motor currents necessary for a wheel start, increasing the current slowly from a start at zero wheelspeed, zero current. The drive motor scale factors needed for these computations were obtained from the startruns: the total generated torque at a certain motor-current value equals the sum of acceleration torque and Coulomb friction value.
The Coulomb friction values were obtained from speed reversal and rundown plots. As an example see the Figs. 6 and 7 for ball bearing wheel TN and Fig. 9 for magnetic bearing wheel EM. The latter clearly shows the absence of Coulomb friction in a magnetic bearing wheel. Also, the magnetic bearing wheels showed a zero stiction value.
Viscous friction values for the wheels at different wheel speeds could be computed from the motor current values necessary for obtaining the different constant wheel speeds for the spectral measurements. Motor gain factor and friction values for the different wheels are given in Table 3. The resulting friction values are plotted in Fig. 10. As shown, the viscous friction of the ball bearing wheels is a non-linear function of wheel speed.

Noise spectra. As can be seen in Figs. 4, 5 and 9 an important noise source is DC motor commutation torque ripple. The ripple value and frequency greatly depends on the number of drive motor pole pairs and phase windings, and on the quality of motor current commutation switching.

The spectral plots given in this paper are all determined at very low motor torque levels i.e. the torque to overcome the total friction at the selected wheel speed. During dynamic operation of the wheel in a control system, larger motor currents will be applied. The commutation peaks in the power spectral density of the wheel torque will increase accordingly.
An example of a spectrum clearly showing commutation noise is shown in Figs. 11 and 12 for magnetic bearing wheel SM.

An AC drive motor, having no commutation, produces a lower noise level (see Fig. 13).

In ball bearing wheels another noise source is present: due to the movements of the retainer, a number of peaks at wheel speed dependent frequencies are generated (r, 2r... where r = cw, the constant c depending on the bearing dimensions). Due to some unknown cause these frequencies all are mixed non-linearly with the wheel speed harmonics, causing peaks down into the low-frequency range. This may be important for noise propagation in satellite attitude control loops.
An example of the noise spectrum generated by a ball bearing wheel (TN) is given in Fig. 14.

A drawback of magnetic wheels for attitude control of a spacecraft can be the existence of nutation frequencies. However, due to the stable wheel support the effect was not observed as such in the wheels studied.

MINIMIZING THE EFFECT OF TORQUE NOISE BY WHEEL STATE FEEDBACK

Introduction

In the remainder of this paper attention will be focussed on possibilities to limit the effect of wheel torque irregularities on spacecraft attitude stability by proper design of the wheel control system.
The increasing power of onboard processors together with the application of modern control techniques allows more freedom in the design of such wheel control systems. It will be shown that performance improvement can be obtained with the use of wheel state feedback.

A three-axis spacecraft control system using reaction wheels can be schematized as shown in Fig. 15. The wheel controller system part of this scheme accepts three orthogonal torque commands from the spacecraft controller, one for each S/C control axis. Functions of such a wheel controller system may include: torque distribution, wheel state feedback, wheel failure detection, autonomous wheel control reconfiguration and wheel momentum unloading. These tasks can be implemented either in a central AOCS computer or in a dedicated processor for the wheel system.

Control Model For The Performance Analysis

For the purpose of studying the basic performance aspects associated with wheel state feedback a single axis model is adopted. Extension of the results towards a three-axis S/C control system is quite straight forward when the torque distribution scheme of Fig. 15 is used. Assuming the wheel torque irregularities of the individual wheels to be mutually uncorrelated, the total wheel induced torque irregularity for a particular spacecraft axis is the root-sum-square of the torque noise of all contributing wheels, weighted according to the orientation of the wheel rotation axes.

There exists a large variety of digital spacecraft control design techniques. Many particular choices have to be made, e.g. sample frequency and quantisation levels, which makes comparison of different systems difficult. Therefore it was decided to evaluate the wheel feedback concept analytically, in the continuous-time domain. State observation and state-feedback is assumed, as basically shown in Fig. 16. PD spacecraft control is assumed. The following treatment focuses on the effect of torque noise on the S/C attitude error. Therefore the S/C state observation process is deleted and the state estimation errors, which include the noise of the sensors, are ignored. The simple S/C control scheme of Fig. 17 remains.

Wheel State Feedback

Three basic types of wheel measurements could be conceived for wheel state feedback: wheel torque, wheel angular velocity or wheel angular attitude. Which of these measurement types will be used, singly or in combination, depends on the available wheel sensors and the wheel control requirements. Most present day wheels are equipped with a pulse counting "tachometer", which in fact is a wheel attitude sensor. In the sequel the use of such an attitude sensor is always assumed. Figure 18 gives the wheel loop block diagram of this wheel attitude measurement case.
Note that the wheel motor is current controlled.
From the torque command T_c, a wheel attitude command α_c must be calculated. This is basically accomplished with a wheel model represented by the transfer function $1/(I_r s^2)$.
This wheel attitude command is compared with the measured wheel attitude α_m to obtain the wheel attitude error α_ε. The wheel controller H_α has the task to minimize this error i.e. the actual wheel is forced to follow the wheel model (model following control). To ease the wheel controller task, a feedforward of T_c will be applied as shown in Fig. 18. The reason for this feedforward path becomes obvious when the disturbances T_i (wheel torque noise) and S_α (wheel sensor noise) are assumed zero. Then the error α_ε will always be zero after a possible initial error is removed by the controller. Therefore this wheel control loop structure has the advantage that the dynamics of the S/C control loop is not affected by the wheel loop, as shown by the transfer function: $T_r/T_c = 1$. The wheel controller is purely dedicated to the task to minimize the effect of the internal torque noise T_i in the total wheel torque T_r. This is not the case when the feedforward path is deleted. In the latter case the transfer function becomes:
$T_r/T_c = (H_\alpha/I_r)/(s^2 + H_\alpha/I_r)$. Here wheel loop dynamics enter in the S/C loop dynamics.
The effect of the torque noise T_i and the

wheel sensor noise S_α on the total wheel torque T_r is given by the transfer functions (assuming a controller of the form: $H_\alpha = P_\alpha + D_\alpha s$):

$$\frac{T_r(s)}{T_i(s)} = \frac{s^2}{s^2+(D_\alpha/I_r)s+P_\alpha/I_r} = \frac{s^2}{s^2+2\zeta_r\omega_{nr}s+\omega_{nr}^2} \quad (1)$$

$$\frac{T_r(s)}{S_\alpha(s)} = \frac{-D_\alpha s^3-P_\alpha s^2}{s^2+(D_\alpha/I_r)s+P_\alpha/I_r} = \frac{-D_\alpha s^3-P_\alpha s^2}{s^2+2\zeta_r\omega_{nr}s+\omega_{nr}^2} \quad (2)$$

where: P_α = wheel attitude error controller gain (Nm/rad)
D_α = wheel velocity error controller gain (Nms/rad)
ζ_r = relative damping of the wheel control loop
ω_{nr} = bandwidth of the wheel control loop (rad/s)

These transfer functions are valid for both the feedforward and the non-feedforward case.

The resulting S/C control system is given in Fig. 19. When this system is compared with the S/C control system without wheel state feedback (Fig. 17), the only difference appears to be the filtering or coloring of the internal disturbance torque T_i and the addition of colored wheel sensor noise. In principle the same S/C controller can be used, thus achieving exactly the same system dynamics as in the case without wheel state feedback. An additional benefit is the possibility to optimize the system for the reduced wheel torque irregularities.

Spacecraft Performance Without Wheel State Feedback

First the performance of the system with a "conventional" current controlled wheel is derived. This result will be used for a performance comparison with systems which include wheel state feedback. The single parameter selected for performance qualification is the variance of the S/C attitude error σ_φ^2, resulting from wheel torque irregularities.

The closed loop performance is analysed with the aid of the following transfer functions (again a PD spacecraft control law is assumed):

$$\frac{\varphi(s)}{\varphi_c(s)} = \frac{(D/I_S)(s+P/D)}{s^2+(D/I_S)s+P/I_S} = \frac{(D/I_S)(s+P/D)}{s^2+2\zeta_S\omega_{nS}s+\omega_{nS}^2} \quad (3)$$

$$\frac{\varphi(s)}{T_i(s)} = \frac{\varphi(s)}{T_e(s)} = \frac{1/I_S}{s^2+(D/I_S)s+P/I_S} \quad (4)$$

WHERE: ζ_S = relative damping of the S/C control loop
ω_{nS} = bandwidth of the S/C control loop (rad/s).

The root locus of the control loop (φ/φ_c) is given in Fig. 20 assuming a constant ratio P/D. The effect of the disturbance torques T_i and T_e on the S/C attitude is characterized by the Bode diagram of Fig. 21.
For the remainder of this analysis, T_i is assumed to be a normally distributed, white noise process, with zero mean and a power spectral density p_T. Note that the average of the internal wheel torque T_i is considered to be removed by the controller set-up shown in Fig. 16. The S/C attitude error due to the wheel torque noise T_i is expressed by the variance (see Prins, 1982).

$$\sigma_\varphi^2 = \frac{p_T}{2PD} = \frac{p_T}{4I_S^2\zeta_S\omega_{nS}^3} \quad (5)$$

Spacecraft Performance With Wheel State Feedback

For the S/C control system with wheel state feedback as given by Figs. 18 and 19, the same performance parameter σ_φ^2 can be derived. The When a PD wheel controller is used the result is (see Prins, 1982):

$$\sigma_\varphi^2 = \frac{p_T}{4I_S^2\zeta_S\omega_{nS}^3} \cdot \frac{b\{(4\zeta_r^2\zeta_S+\zeta_S)a^5+(16\zeta_r^3\zeta_S^2+\zeta_r)a^4+16\zeta_r^4\zeta_S a^3+4\zeta_r^3 a^2\}+\zeta_S a+\zeta_r}{\zeta_r\{a^4+4\zeta_r\zeta_S a^3+2(2\zeta_r^2+2\zeta_S^2-1)a^2+4\zeta_r\zeta_S a+1\}} \quad (6)$$

where:
$a = \omega_{nr}/\omega_{nS}$, bandwidth ratio of the wheel loop and the S/C control loop
$b = I_r^2\omega_{nS}^4\, p_\alpha/p_T$, proportional to the ratio of the PSD of the wheel sensor noise p_α and the wheel torque noise p_T.

The same analysis can be done for a velocity-only wheel feedback controller, $H_\alpha = D_\alpha s$. This represents the case of wheel velocity control with the use of the wheel attitude sensor. Fleming and Ramos (1979) report on the application of a similar case, however, without the feedworward path presented in this paper. The resulting S/C attitude variance is (see Prins, 1982):

$$\sigma_\varphi^2 = \frac{p_T}{4I_S^2\zeta_S\omega_{nS}^3} \quad \frac{b\{2\zeta_S a^3+a^2\}+1}{a^2+2\zeta_S a+1} \quad (7)$$

Noise Reduction Factor

The first factor of Eqs. (6) and (7) are identical to the variance derived for the case without wheel state feedback, Eq. (5). The possible improvement due to wheel state feedback can be described by the noise reduction factor F, defined by:

$$F = \frac{\sigma_\varphi^2 \text{ with wheel state feedback}}{\sigma_\varphi^2 \text{ without wheel state feedback}} \qquad (8)$$

F is given by the second factor in Eqs. (6) and (7) and is plotted in Figs. 22 and 23 respectively as a function of a and b for ζ_S $(= \zeta_r) = 0.7$. From these figures a number of observations can be made:

- No wheel state feedback means a = 0 and therefore the noise reduction is zero as expected: F = 1.
- There exists an optimal bandwidth ratio (a) for given values of $\zeta_s(,\zeta_r)$ and b, i.e. for a given S/C controller bandwidth ω_{nS}, a bandwidth of the wheel state feedback loop ω_{nr} can be found which gives a maximum noise reduction (F minimal).
- Whether wheel state feedback gives a better S/C control performance, heavily depends on the parameter b. For a smaller value of b wheel state feedback, gives a larger noise reduction for any bandwidth ratio a.
- The correct choice of the ratio of wheel loop bandwidth to S/C control loop bandwidth (a) is very important. When too high a wheel loop bandwidth is chosen, the application of wheel state feedback may even increase the S/C attitude error (F>1). This is due to the effect of wheel sensor noise.
- The curve b=0 represents the ideal case where the wheel sensor noise is zero ($p_\alpha = 0$).

  The effect of wheel torque noise reduces to zero for sufficiently high wheel loop bandwidth. The noise added by the wheel sensor noise is represented by the difference between any b-curve and the curve for b = 0.

<u>Comparison Of Results</u>

Comparison of Figs. 22 and 23 shows that for a given b, the D_α wheel controller gives in most cases a better noise reduction than the $P_\alpha D_\alpha$ controller.

A disadvantage of the D_α controller, however, is its non-zero wheel velocity error in the presence of a low frequency or fixed internal disturbance torque T_i. The $P_\alpha D_\alpha$ wheel controller has a zero wheel velocity error for fixed T_i, which has a favorable effect on the S/C attitude and velocity errors in situations where the mean level of T_i suddenly changes (e.g. a wheel-speed reversal in a ball bearing wheel system).

The S/C controller bandwidth is always the result of a compromise between steady state performance without significant T_i changes and the transient behaviour with more rapid disturbance torque changes. The $P_\alpha D_\alpha$ wheel controller possibly allows a lower S/C controller bandwidth ω_{nS}, since it counteracts T_i changes more efficiently due to its zero velocity error. As the value of b varies with the fourth power of ω_{nS}, a reduction in ω_{nS} gives a considerable reduction in F. Also the S/C attitude error due to S/C sensor noise decreases with ω_{nS}. The conclusion is that due to the decreased S/C control bandwidth, the $P_\alpha D_\alpha$ wheel controller may provide a better noise reduction than the D_α wheel controller in a practical design case.

CONCLUSIONS

Torque and force irregularities of rotating reaction wheels result in discrete wheel speed dependent spectral peaks. An important source is the motor current commutation as present in d.c. wheel drive motors; its frequency (located at higher wheel speed harmonics) is depending on the number of motor pole pairs and phase windings. Its amplitude depends on current magnitude and commutation quality.

In ball bearings another noise source is present due to the retainer movements, resulting in peaks at the retainer speed and its harmonics. These frequencies are mixed nonlinearly with the wheel speed harmonics, resulting in peaks down into the low-frequency range.

Poorly damped natural frequencies present in the magnet bearing passively controlled DOF may cause an amplification of noise when excited by wheel noise peaks.

The results of the first part of the study did not prove the total amount of noise in magnetic bearing wheels to be substantially below that of ball bearing wheels.

Reaction wheel state feedback, as discussed in the last part of this paper, can give a considerable spacecraft control performance improvement. Further details on this subject can be found in Prins (1982). In this reference digital implementation aspects are treated as well. Computer simulations were outside the scope of this limited study.

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Reference	TN	TA	TD	EM	TM	SM
Manufacturer	TELDIX	TELDIX	TELDIX	ESTEC	TELDIX	SNIAS
Type	RSR 14	RSR 2-2	RSR 2-0	MBRW 2	MRR 2	RRPM 2
Diameter (mm)	350	205	205	200	216	350
Height (mm)	120	75	75	73	87	150
Mass (kg)	6.3	2.75	2.6	2.5	3	7.2
Mass rotor (kg)	3.21	0.89	0.78	1.0	0.87	3.61
Inertia rotor (kgm^2)	0.0716	0.0067	0.0054	0.0076	0.0064	0.0600
Max. momentum (Nms)	$\pm$ 14	$\pm$ 1.86	$\pm$ 2	$\pm$ 2	$\pm$ 2	$\pm$ 15
Max. speed (rpm)	$\overline{2}000$	$\overline{2}450$	$\overline{3}500$	$\overline{2}500$	$\overline{3}200$	$\overline{2}400$
Max. torque (Nm)	0.2	0.03	$\pm$ 0.1	$\pm$ 0.05	$\pm$ 0.1	0.2
Type of bearings	ball	ball	ball	magnetic	magnetic	magnetic
Type of motor	d.c.	a.c.	d.c.	d.c.	d.c.	d.c.
Number of pole pairs	8	8	6	10	20	18
Number of phases	3	2	3	3	3	4
Active controlled axes	-	-	-	radial (x and y)	radial (x and y)	axial (z)
Commutation sensor	prox. osc.	optical	prox. osc.	reflective optical	prox. osc.	prox. osc.

TABLE 1 Characteristics of Tested Reaction Wheels

	Reaction wheels					
	ballbearing			magnetbearing		
	TN	TA	TD	EM	TM	SM
wheel-speeds (rpm)	44	50	100	100	99	110
	-100	-100	-550	-528	-528	-550
	521	600	1056	1100	1089	1080
	2045	2150	2660	2600	2121	2130

TABLE 2 Selected Wheelspeeds for Spectral Runs

[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]		
					[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Note: Coulomb friction and stiction values for the magnetic bearing wheels are near zero (not measurable)

TABLE 3 Approximate Measured Motor and Friction Constants

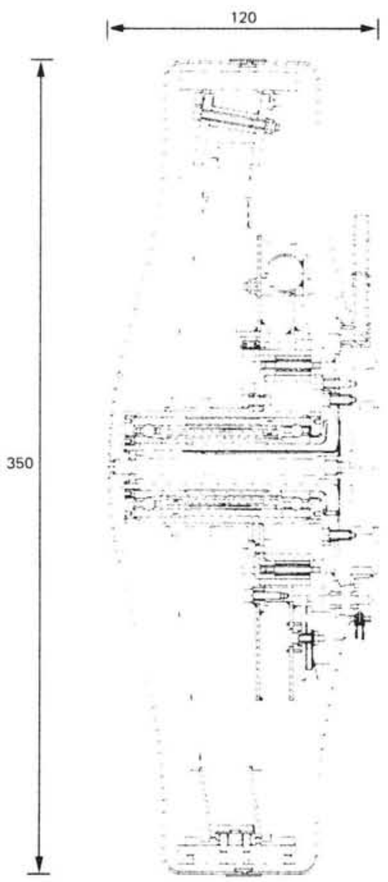

Fig. 1 Reaction wheel TN (Teldix Type RSR 14)

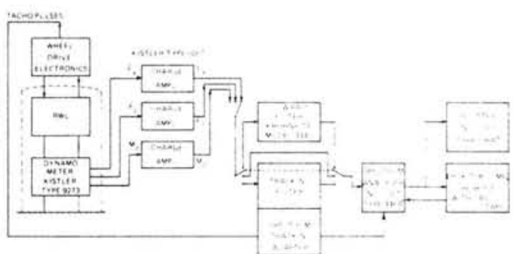

Fig. 2 Data acquisition and processing for reaction-wheel noise measurements

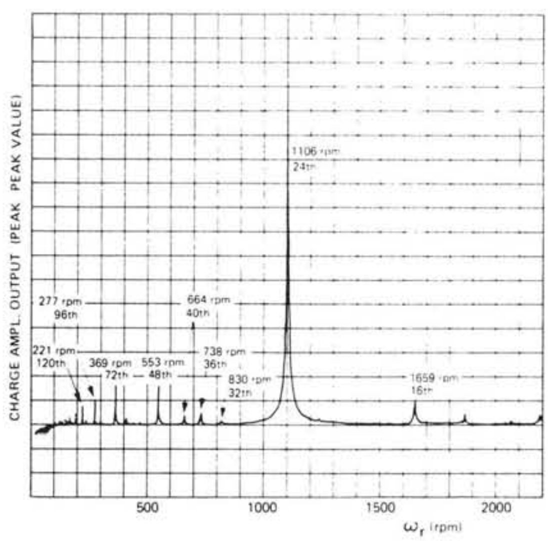

Fig. 3 RWL-TN, M_z channel, search run

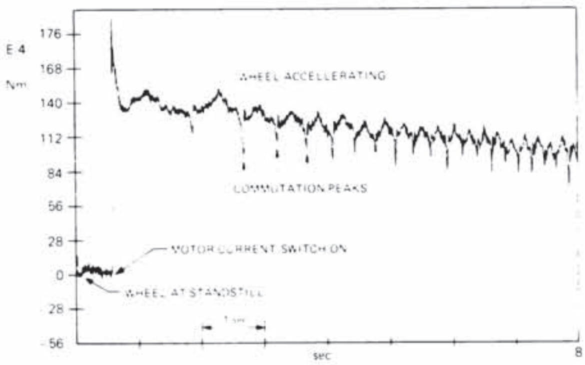

Fig. 4 RWL-TN, M_z channel, startrun at 200 mA motor current

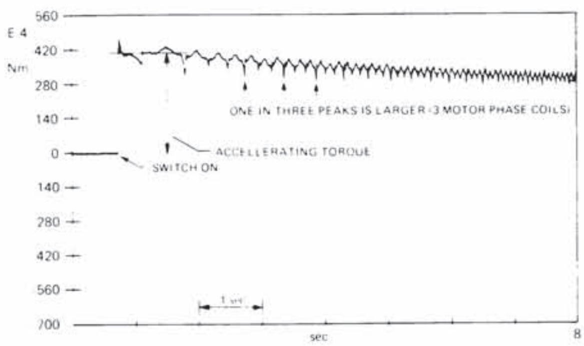

Fig. 5 RWL-TN, M_z channel, startrun at 500 mA motor current

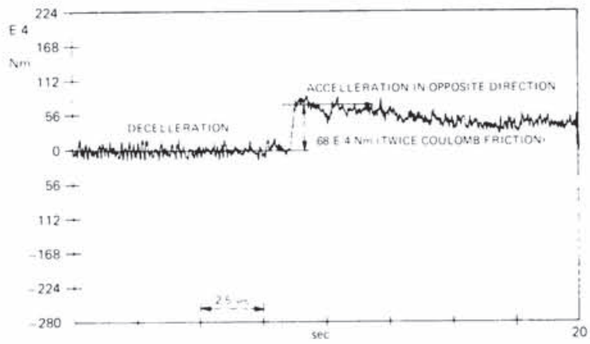

Fig. 6 RWL-TN, M_z channel, speed reversal at 200 mA^z motor current

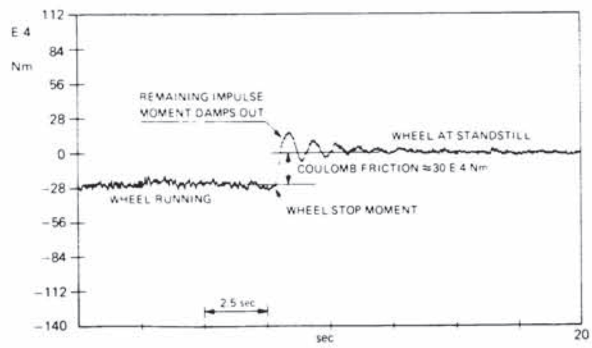

Fig. 7 RWL-TN, M_z channel, rundown test

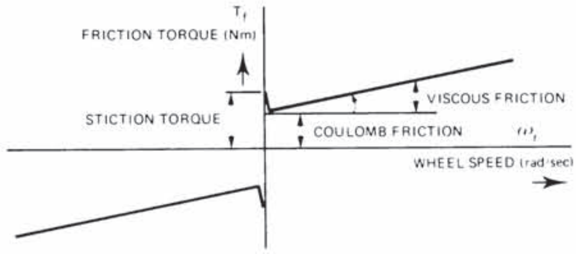

Fig. 8 Simplified reaction wheel friction model

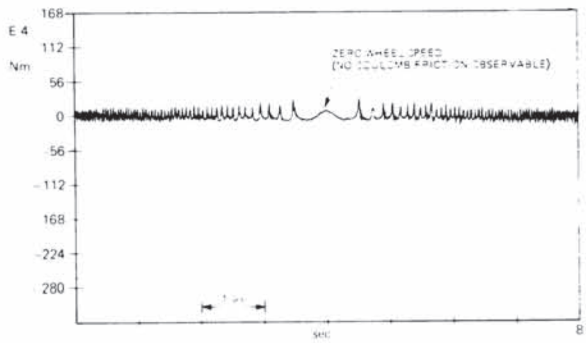

Fig. 9 RWL-EM, M_z channel, speed reversal at 500 mA

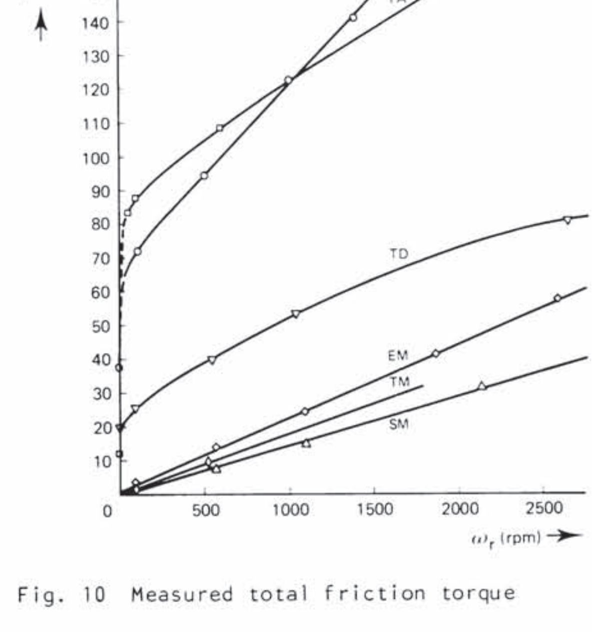

Fig. 10 Measured total friction torque

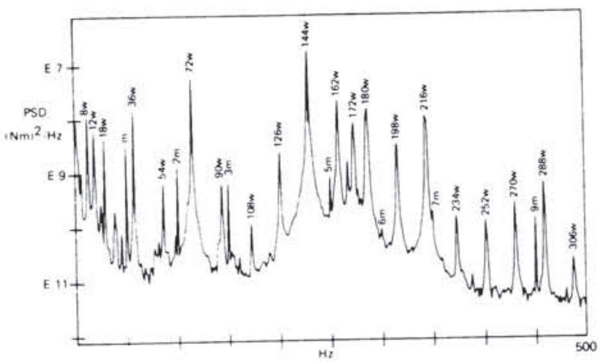

Fig. 11 RWL-SM, M_z channel, power spectrum, -100 rpm, total noise 812E-6 Nm RMS

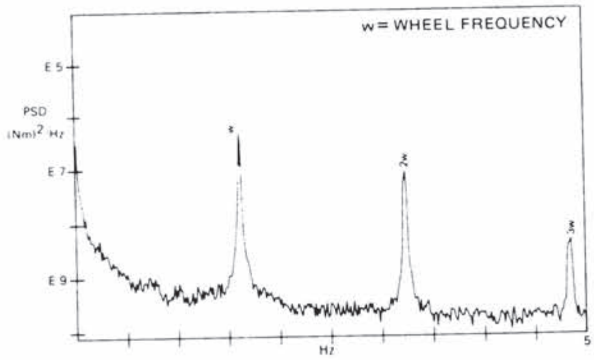

Fig. 12 RWL-SM, M_z channel, power spectrum, -100 rpm, total noise 155E-6 Nm RMS

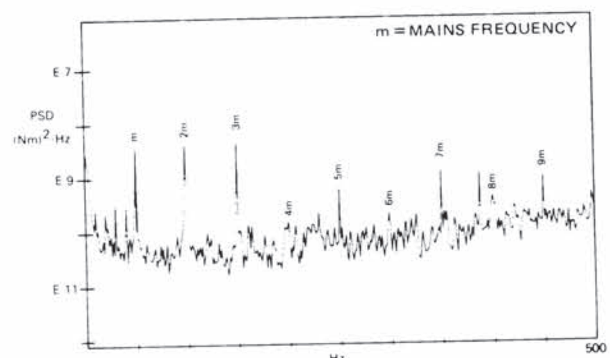

Fig. 13 RWL-TA, M_z channel, power spectrum, -100 rpm, total noise 278E-6 Nm RMS

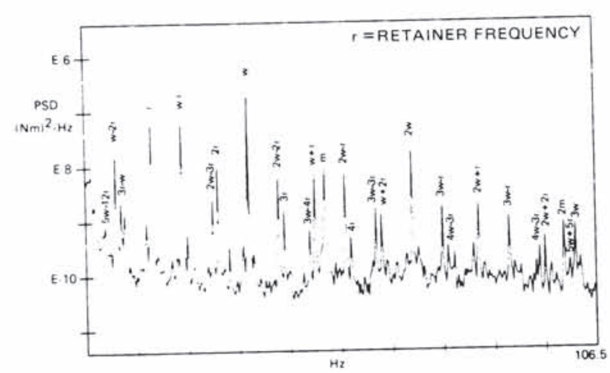

Fig. 14 RWL-TN, M_z channel, power spectrum, 2000 rpm, total noise 442E-6 Nm RMS

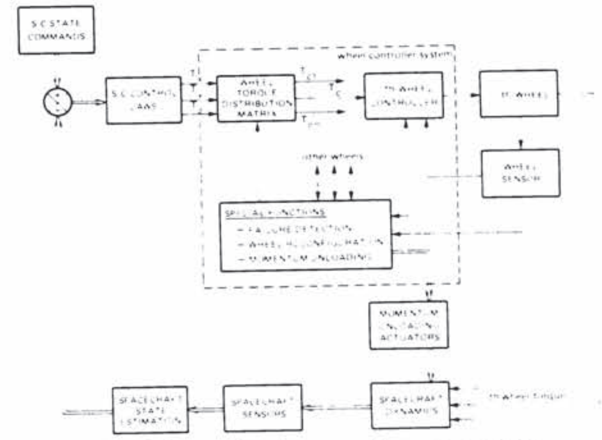

Fig. 15 Schematic three-axis spacecraft control system using reaction wheels

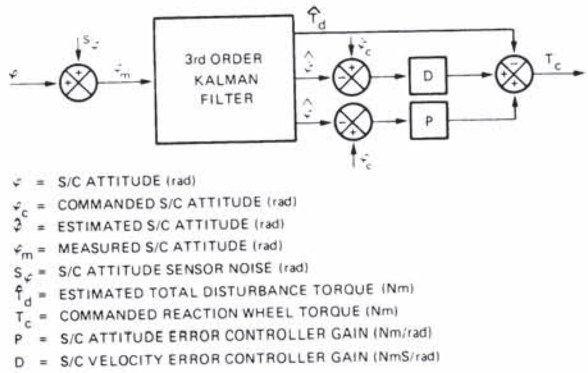

Fig. 16 Spacecraft control process including state observation

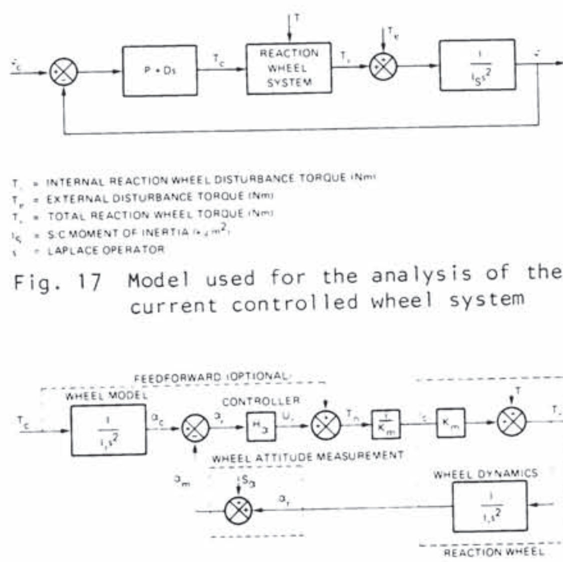

Fig. 17 Model used for the analysis of the current controlled wheel system

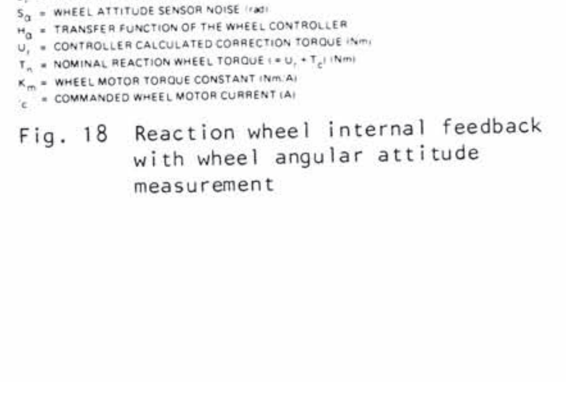

Fig. 18 Reaction wheel internal feedback with wheel angular attitude measurement

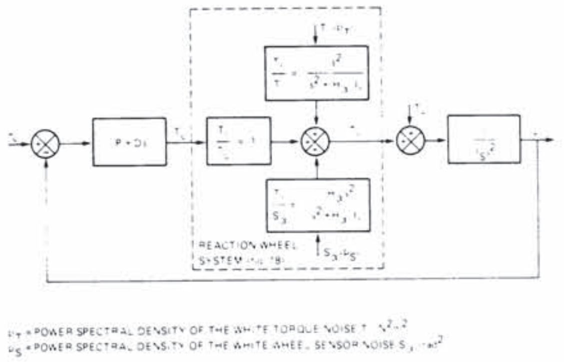

Fig. 19 Single axis S/C control system with a reaction wheel with internal feedback and wheel attitude measurement

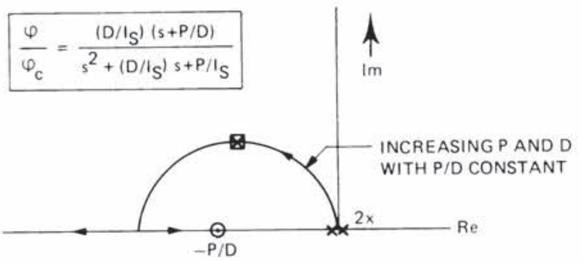

Fig. 20 Rootlocus of the S/C control system

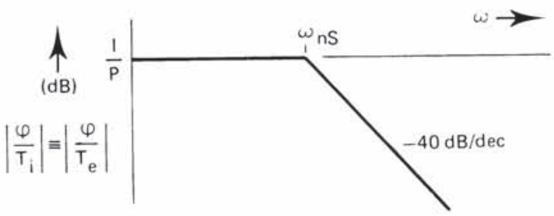

Fig. 21 Bode diagram (asymptotic) for the S/C control system

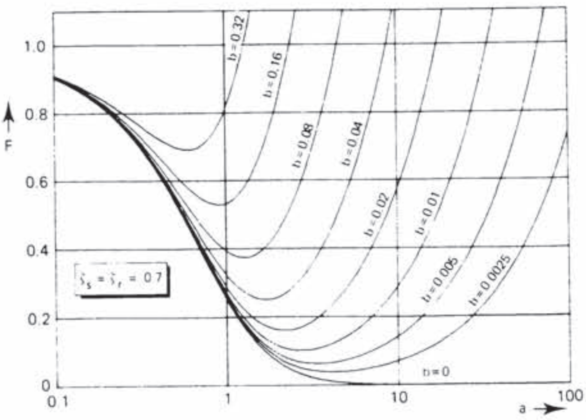

Fig. 22 S/C attitude noise variance reduction factor F due to wheel state feedback (P_α D_α) as a function of a, b, ζ_s and ζ_r

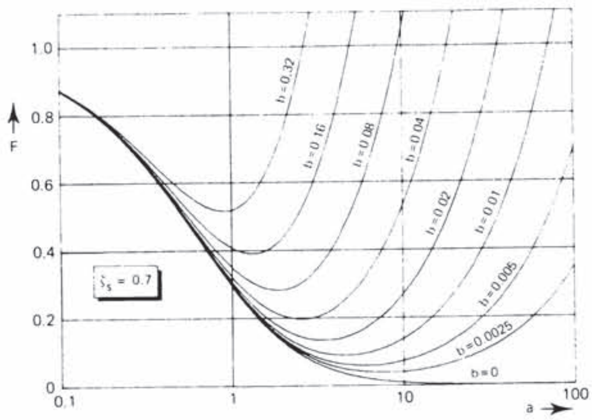

Fig. 23 S/C attitude noise variance reduction factor F due to wheel state feedback (D_α) as a function of a, b and ζ_s